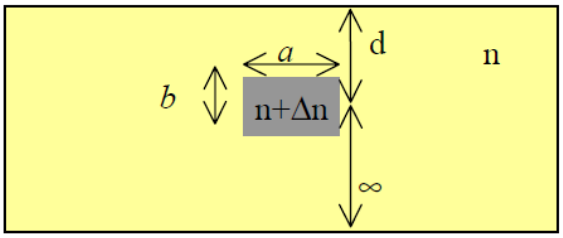
MASTER SISE/PSRS
Introduction to Guided Optics
Homework n° 1

The aim of this exercise is to design a single-mode planar waveguide using a technology that makes it possible to manufacture buried step index waveguides of limited width such as that illustrated in figure 1.
You have to determine the manufacturing parameters so that this guide is single mode between 1260 and 1560nm. To be efficient, this guide must have a minimum of losses and must be compatible with the single mode fiber SMF28. Due to the low Δn, we will not distinguish the case TE from the case TM.

I. Waveguide parameters.

The buried optical waveguide obtained by the used technology is shown in figure 1.

- a = **to be determined**
- d = **to be determined**
- b=5µm
- n=1.46 and Δn=0.003



NB: d = distance between la surface and the center of the waveguide

a) Transverse dimensions of the waveguide.

The waveguide must be single mode. The transverse dimensions of the waveguides must therefore be determined in order to satisfy this unimodal guiding condition. In order to simplify the study of this guide with limited width, we will reduce the study to that of two symmetrical planar guides through the effective index method. We will study the vertical dimension illustrated below first, then the horizontal dimension.

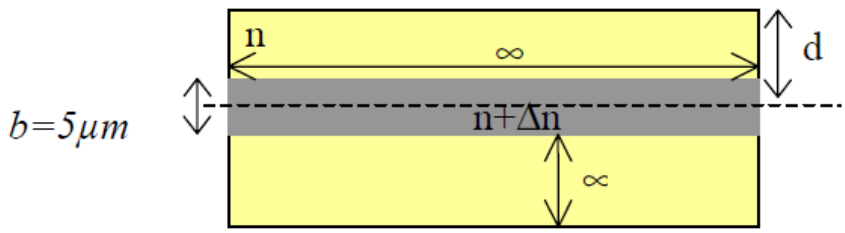


Figure 2: Planar waveguideguide to be studied for the vertical dimension.

1) Is this guide single-mode at the required wavelengths in the vertical direction?
2) Determine the cutoff wavelength of the first higher order mode in the vertical direction.
3) Show that from the effective index method, the study of the guide with limited width can be reduced to the study of the planar waveguide of width *a*, cladding index 1.46 and core index: nc = 1.4616 for λ=1260nm, nc = 1.4613 for λ=1560nm. You will use for this the normalized frequency V and the parameters U and W seen during the lessons (with the TD formalism). (See Annex A).
4) Determine the optimum width *a* (we will take an integer value in µm) so that the guide is single-mode at the required wavelengths with the best possible confinement.
5) Determine in this case the effective indices of the modes guided at 1260nm and 1560nm.

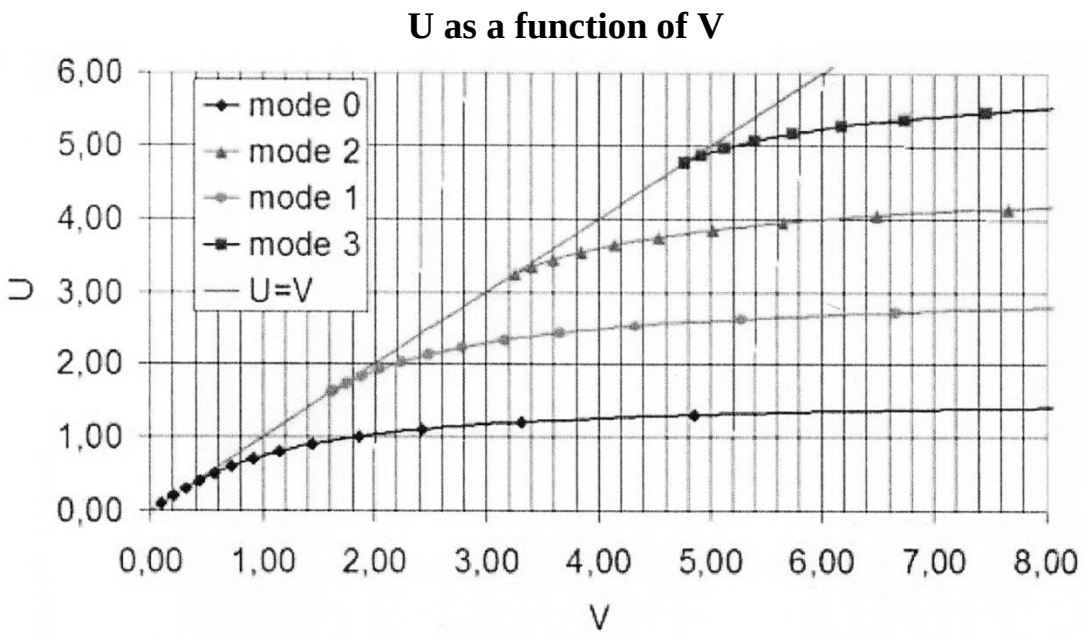
b) Depth of burial of the optical waveguide (vertical direction).
The waveguide will have low losses if the guided electromagnetic field does not see the surface of the optical chip. The waveguides must then be buried to a depth sufficient for this purpose. It will be considered that the electric field no longer sees the surface of the chip when its amplitude at the surface is less than **1% of its maximum amplitude**.
6) Explain why the guided electromagnetic field should not see the surface of the optical chip to have good optical performance. (there are 2 important reasons)
7) Give the solutions of the electric field in a symmetrical planar waveguide for even modes.
8) From these solutions of the electric field, determine the burying depth d (distance between the surface and the center of the guide) to properly isolate the electric field from the surface of the optical chip at 1260nm and 1560nm. Deduce that a burying depth of 20µm allows to have a good optical isolation from the surface.

II. Optical performances.

9) We now want to characterize this waveguide: propagation losses and coupling losses. Propose an optical bench for the measurements of these losses, by explaining the measurements obtained as well as their interpretation.

10) From the field maps given in Appendix B and the specifications given in Appendix C, determine the coupling losses with a SMF28 fiber at 1310nm and 1550 nm.
You will express your results in dB. To calculate these losses, we will use the Gaussian approximation. What do you think of these coupling losses?

APPENDIX A



Resulats de l'équation de dispersion pour un guide plan symétrique en TE/TM

Mode 0			##	Mode 0			Mode 0			Mode 0			Mode 0		
V	U	W		V	U	W	V	U	W	V	U	W	V	U	W
0,9000	0,6926	0,5747		0,9384	0,7110	0,6124	0,9783	0,7294	0,6520	1,0199	0,7478	0,6936	1,0634	0,7662	0,7373
0,9008	0,6930	0,5755		0,9392	0,7114	0,6132	0,9792	0,7298	0,6528	1,0209	0,7482	0,6945	1,0643	0,7666	0,7383
0,9016	0,6934	0,5763		0,9401	0,7118	0,6140	0,9801	0,7302	0,6537	1,0218	0,7486	0,6954	1,0653	0,7670	0,7393
0,9024	0,6938	0,5771		0,9409	0,7122	0,6149	0,9810	0,7306	0,6546	1,0227	0,7490	0,6964	1,0663	0,7674	0,7403
0,9032	0,6942	0,5779		0,9418	0,7126	0,6157	0,9819	0,7310	0,6555	1,0236	0,7494	0,6973	1,0672	0,7678	0,7412
0,9041	0,6946	0,5787		0,9426	0,7130	0,6166	0,9827	0,7314	0,6564	1,0246	0,7498	0,6982	1,0682	0,7682	0,7422
0,9049	0,6950	0,5795		0,9435	0,7134	0,6174	0,9836	0,7318	0,6573	1,0255	0,7502	0,6992	1,0692	0,7686	0,7432
0,9057	0,6954	0,5803		0,9443	0,7138	0,6183	0,9845	0,7322	0,6582	1,0264	0,7506	0,7001	1,0701	0,7690	0,7442
0,9065	0,6958	0,5811		0,9452	0,7142	0,6191	0,9854	0,7326	0,6591	1,0274	0,7510	0,7010	1,0711	0,7694	0,7452
0,9074	0,6962	0,5819		0,9460	0,7146	0,6200	0,9863	0,7330	0,6599	1,0283	0,7514	0,7020	1,0721	0,7698	0,7462
0,9082	0,6966	0,5827		0,9469	0,7150	0,6208	0,9872	0,7334	0,6608	1,0292	0,7518	0,7029	1,0730	0,7702	0,7471
0,9090	0,6970	0,5835		0,9478	0,7154	0,6217	0,9881	0,7338	0,6617	1,0301	0,7522	0,7038	1,0740	0,7706	0,7481
0,9098	0,6974	0,5843		0,9486	0,7158	0,6225	0,9890	0,7342	0,6626	1,0311	0,7526	0,7048	1,0750	0,7710	0,7491
0,9107	0,6978	0,5851		0,9495	0,7162	0,6234	0,9899	0,7346	0,6635	1,0320	0,7530	0,7057	1,0760	0,7714	0,7501
0,9115	0,6982	0,5859		0,9503	0,7166	0,6242	0,9908	0,7350	0,6644	1,0330	0,7534	0,7067	1,0769	0,7718	0,7511
0,9123	0,6986	0,5868		0,9512	0,7170	0,6251	0,9917	0,7354	0,6653	1,0339	0,7538	0,7076	1,0779	0,7722	0,7521
0,9131	0,6990	0,5876		0,9521	0,7174	0,6259	0,9926	0,7358	0,6662	1,0348	0,7542	0,7086	1,0789	0,7726	0,7531
0,9140	0,6994	0,5884		0,9529	0,7178	0,6268	0,9935	0,7362	0,6671	1,0358	0,7546	0,7095	1,0799	0,7730	0,7541
0,9148	0,6998	0,5892		0,9538	0,7182	0,6276	0,9944	0,7366	0,6680	1,0367	0,7550	0,7104	1,0809	0,7734	0,7551
0,9156	0,7002	0,5900		0,9547	0,7186	0,6285	0,9953	0,7370	0,6689	1,0376	0,7554	0,7114	1,0818	0,7738	0,7561
0,9165	0,7006	0,5908		0,9555	0,7190	0,6293	0,9962	0,7374	0,6698	1,0386	0,7558	0,7123	1,0828	0,7742	0,7571
0,9173	0,7010	0,5916		0,9564	0,7194	0,6302	0,9971	0,7378	0,6707	1,0395	0,7562	0,7133	1,0838	0,7746	0,7580
0,9181	0,7014	0,5925		0,9573	0,7198	0,6311	0,9980	0,7382	0,6716	1,0405	0,7566	0,7142	1,0848	0,7750	0,7590
0,9190	0,7018	0,5933		0,9581	0,7202	0,6319	0,9989	0,7386	0,6725	1,0414	0,7570	0,7152	1,0858	0,7754	0,7600
0,9198	0,7022	0,5941		0,9590	0,7206	0,6328	0,9998	0,7390	0,6734	1,0424	0,7574	0,7161	1,0868	0,7758	0,7610
0,9206	0,7026	0,5949		0,9599	0,7210	0,6336	1,0007	0,7394	0,6743	1,0433	0,7578	0,7171	1,0878	0,7762	0,7621
0,9215	0,7030	0,5957		0,9607	0,7214	0,6345	1,0016	0,7398	0,6752	1,0442	0,7582	0,7180	1,0887	0,7766	0,7631
0,9223	0,7034	0,5966		0,9616	0,7218	0,6354	1,0025	0,7402	0,6761	1,0452	0,7586	0,7190	1,0897	0,7770	0,7641
0,9232	0,7038	0,5974		0,9625	0,7222	0,6362	1,0034	0,7406	0,6770	1,0461	0,7590	0,7199	1,0907	0,7774	0,7651
0,9240	0,7042	0,5982		0,9634	0,7226	0,6371	1,0043	0,7410	0,6780	1,0471	0,7594	0,7209	1,0917	0,7778	0,7661
0,9248	0,7046	0,5990		0,9642	0,7230	0,6380	1,0053	0,7414	0,6789	1,0480	0,7598	0,7219	1,0927	0,7782	0,7671
0,9257	0,7050	0,5999		0,9651	0,7234	0,6388	1,0062	0,7418	0,6798	1,0490	0,7602	0,7228	1,0937	0,7786	0,7681
0,9265	0,7054	0,6007		0,9660	0,7238	0,6397	1,0071	0,7422	0,6807	1,0499	0,7606	0,7238	1,0947	0,7790	0,7691
0,9274	0,7058	0,6015		0,9669	0,7242	0,6406	1,0080	0,7426	0,6816	1,0509	0,7610	0,7247	1,0957	0,7794	0,7701
0,9282	0,7062	0,6023		0,9677	0,7246	0,6414	1,0089	0,7430	0,6825	1,0518	0,7614	0,7257	1,0967	0,7798	0,7711
0,9290	0,7066	0,6032		0,9686	0,7250	0,6423	1,0098	0,7434	0,6834	1,0528	0,7618	0,7267	1,0977	0,7802	0,7721
0,9299	0,7070	0,6040		0,9695	0,7254	0,6432	1,0107	0,7438	0,6844	1,0538	0,7622	0,7276	1,0987	0,7806	0,7731
0,9307	0,7074	0,6048		0,9704	0,7258	0,6441	1,0116	0,7442	0,6853	1,0547	0,7626	0,7286	1,0997	0,7810	0,7742
0,9316	0,7078	0,6057		0,9712	0,7262	0,6449	1,0126	0,7446	0,6862	1,0557	0,7630	0,7296	1,1007	0,7814	0,7752
0,9324	0,7082	0,6065		0,9721	0,7266	0,6458	1,0135	0,7450	0,6871	1,0566	0,7634	0,7305	1,1017	0,7818	0,7762
0,9333	0,7086	0,6073		0,9730	0,7270	0,6467	1,0144	0,7454	0,6880	1,0576	0,7638	0,7315	1,1027	0,7822	0,7772
0,9341	0,7090	0,6082		0,9739	0,7274	0,6476	1,0153	0,7458	0,6890	1,0585	0,7642	0,7325	1,1037	0,7826	0,7782
0,9350	0,7094	0,6090		0,9748	0,7278	0,6484	1,0162	0,7462	0,6899	1,0595	0,7646	0,7334	1,1047	0,7830	0,7793
0,9358	0,7098	0,6098		0,9757	0,7282	0,6493	1,0172	0,7466	0,6908	1,0605	0,7650	0,7344	1,1057	0,7834	0,7803
0,9367	0,7102	0,6107		0,9765	0,7286	0,6502	1,0181	0,7470	0,6917	1,0614	0,7654	0,7354	1,1067	0,7838	0,7813
0,9375	0,7106	0,6115		0,9774	0,7290	0,6511	1,0190	0,7474	0,6927	1,0624	0,7658	0,7364	1,1077	0,7842	0,7823

Resulats de l'équation de dispersion pour un guide plan symétrique en TE/TM

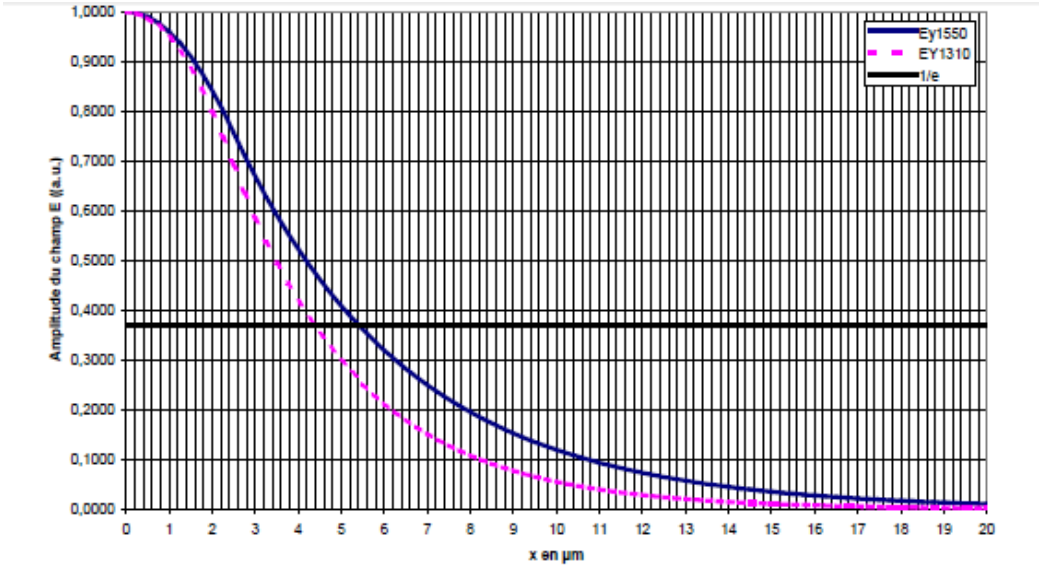
Mode 0			Mode 0			Mode 0			Mode 0			Mode 0		
V	U	W	V	U	W	V	U	W	V	U	W	V	U	W
1,1087	0,7846	0,7833	1,1561	0,8030	0,8318	1,2058	0,8214	0,8828	1,2579	0,8398	0,9365	1,3126	0,8582	0,9932
1,1097	0,7850	0,7844	1,1572	0,8034	0,8329	1,2069	0,8218	0,8839	1,2591	0,8402	0,9377	1,3139	0,8586	0,9945
1,1107	0,7854	0,7854	1,1583	0,8038	0,8339	1,2080	0,8222	0,8851	1,2602	0,8406	0,9389	1,3151	0,8590	0,9958
1,1117	0,7858	0,7864	1,1593	0,8042	0,8350	1,2091	0,8226	0,8862	1,2614	0,8410	0,9401	1,3163	0,8594	0,9970
1,1127	0,7862	0,7875	1,1604	0,8046	0,8361	1,2103	0,8230	0,8873	1,2626	0,8414	0,9413	1,3175	0,8598	0,9983
1,1138	0,7866	0,7885	1,1614	0,8050	0,8372	1,2114	0,8234	0,8885	1,2637	0,8418	0,9425	1,3187	0,8602	0,9996
1,1148	0,7870	0,7895	1,1625	0,8054	0,8383	1,2125	0,8238	0,8896	1,2649	0,8422	0,9438	1,3200	0,8606	1,0008
1,1158	0,7874	0,7906	1,1636	0,8058	0,8394	1,2136	0,8242	0,8908	1,2661	0,8426	0,9450	1,3212	0,8610	1,0021
1,1168	0,7878	0,7916	1,1646	0,8062	0,8405	1,2147	0,8246	0,8919	1,2672	0,8430	0,9462	1,3224	0,8614	1,0034
1,1178	0,7882	0,7926	1,1657	0,8066	0,8415	1,2158	0,8250	0,8931	1,2684	0,8434	0,9474	1,3237	0,8618	1,0047
1,1188	0,7886	0,7937	1,1667	0,8070	0,8426	1,2169	0,8254	0,8942	1,2696	0,8438	0,9486	1,3249	0,8622	1,0060
1,1199	0,7890	0,7947	1,1678	0,8074	0,8437	1,2180	0,8258	0,8954	1,2708	0,8442	0,9498	1,3261	0,8626	1,0072
1,1209	0,7894	0,7957	1,1689	0,8078	0,8448	1,2192	0,8262	0,8965	1,2719	0,8446	0,9510	1,3274	0,8630	1,0085
1,1219	0,7898	0,7968	1,1699	0,8082	0,8459	1,2203	0,8266	0,8977	1,2731	0,8450	0,9522	1,3286	0,8634	1,0098
1,1229	0,7902	0,7978	1,1710	0,8086	0,8470	1,2214	0,8270	0,8988	1,2743	0,8454	0,9535	1,3298	0,8638	1,0111
1,1239	0,7906	0,7989	1,1721	0,8090	0,8481	1,2225	0,8274	0,9000	1,2755	0,8458	0,9547	1,3311	0,8642	1,0124
1,1250	0,7910	0,7999	1,1732	0,8094	0,8492	1,2237	0,8278	0,9012	1,2766	0,8462	0,9559	1,3323	0,8646	1,0137
1,1260	0,7914	0,8010	1,1742	0,8098	0,8503	1,2248	0,8282	0,9023	1,2778	0,8466	0,9571	1,3336	0,8650	1,0150
1,1270	0,7918	0,8020	1,1753	0,8102	0,8514	1,2259	0,8286	0,9035	1,2790	0,8470	0,9584	1,3348	0,8654	1,0163
1,1280	0,7922	0,8031	1,1764	0,8106	0,8525	1,2270	0,8290	0,9046	1,2802	0,8474	0,9596	1,3360	0,8658	1,0176
1,1291	0,7926	0,8041	1,1775	0,8110	0,8536	1,2282	0,8294	0,9058	1,2814	0,8478	0,9608	1,3373	0,8662	1,0188
1,1301	0,7930	0,8051	1,1785	0,8114	0,8547	1,2293	0,8298	0,9070	1,2826	0,8482	0,9620	1,3385	0,8666	1,0201
1,1311	0,7934	0,8062	1,1796	0,8118	0,8558	1,2304	0,8302	0,9081	1,2837	0,8486	0,9633	1,3398	0,8670	1,0214
1,1322	0,7938	0,8073	1,1807	0,8122	0,8569	1,2316	0,8306	0,9093	1,2849	0,8490	0,9645	1,3410	0,8674	1,0227
1,1332	0,7942	0,8083	1,1818	0,8126	0,8581	1,2327	0,8310	0,9105	1,2861	0,8494	0,9657	1,3423	0,8678	1,0240
1,1342	0,7946	0,8094	1,1829	0,8130	0,8592	1,2338	0,8314	0,9116	1,2873	0,8498	0,9670	1,3435	0,8682	1,0253
1,1353	0,7950	0,8104	1,1839	0,8134	0,8603	1,2350	0,8318	0,9128	1,2885	0,8502	0,9682	1,3448	0,8686	1,0267
1,1363	0,7954	0,8115	1,1850	0,8138	0,8614	1,2361	0,8322	0,9140	1,2897	0,8506	0,9694	1,3461	0,8690	1,0280
1,1373	0,7958	0,8125	1,1861	0,8142	0,8625	1,2372	0,8326	0,9152	1,2909	0,8510	0,9707	1,3473	0,8694	1,0293
1,1384	0,7962	0,8136	1,1872	0,8146	0,8636	1,2384	0,8330	0,9163	1,2921	0,8514	0,9719	1,3486	0,8698	1,0306
1,1394	0,7966	0,8146	1,1883	0,8150	0,8647	1,2395	0,8334	0,9175	1,2933	0,8518	0,9732	1,3498	0,8702	1,0319
1,1404	0,7970	0,8157	1,1894	0,8154	0,8659	1,2407	0,8338	0,9187	1,2945	0,8522	0,9744	1,3511	0,8706	1,0332
1,1415	0,7974	0,8168	1,1905	0,8158	0,8670	1,2418	0,8342	0,9199	1,2957	0,8526	0,9756	1,3524	0,8710	1,0345
1,1425	0,7978	0,8178	1,1915	0,8162	0,8681	1,2429	0,8346	0,9211	1,2969	0,8530	0,9769	1,3536	0,8714	1,0358
1,1436	0,7982	0,8189	1,1926	0,8166	0,8692	1,2441	0,8350	0,9222	1,2981	0,8534	0,9781	1,3549	0,8718	1,0371
1,1446	0,7986	0,8200	1,1937	0,8170	0,8703	1,2452	0,8354	0,9234	1,2993	0,8538	0,9794	1,3562	0,8722	1,0385
1,1456	0,7990	0,8210	1,1948	0,8174	0,8715	1,2464	0,8358	0,9246	1,3005	0,8542	0,9806	1,3574	0,8726	1,0398
1,1467	0,7994	0,8221	1,1959	0,8178	0,8726	1,2475	0,8362	0,9258	1,3017	0,8546	0,9819	1,3587	0,8730	1,0411
1,1477	0,7998	0,8232	1,1970	0,8182	0,8737	1,2487	0,8366	0,9270	1,3029	0,8550	0,9831	1,3600	0,8734	1,0424
1,1488	0,8002	0,8242	1,1981	0,8186	0,8748	1,2498	0,8370	0,9282	1,3041	0,8554	0,9844	1,3612	0,8738	1,0438
1,1498	0,8006	0,8253	1,1992	0,8190	0,8760	1,2510	0,8374	0,9294	1,3053	0,8558	0,9857	1,3625	0,8742	1,0451
1,1509	0,8010	0,8264	1,2003	0,8194	0,8771	1,2521	0,8378	0,9305	1,3066	0,8562	0,9869	1,3638	0,8746	1,0464
1,1519	0,8014	0,8275	1,2014	0,8198	0,8782	1,2533	0,8382	0,9317	1,3078	0,8566	0,9882	1,3651	0,8750	1,0477
1,1530	0,8018	0,8285	1,2025	0,8202	0,8794	1,2544	0,8386	0,9329	1,3090	0,8570	0,9894	1,3663	0,8754	1,0491
1,1540	0,8022	0,8296	1,2036	0,8206	0,8805	1,2556	0,8390	0,9341	1,3102	0,8574	0,9907	1,3676	0,8758	1,0504
1,1551	0,8026	0,8307	1,2047	0,8210	0,8816	1,2568	0,8394	0,9353	1,3114	0,8578	0,9920	1,3689	0,8762	1,0517

Suite annexe A

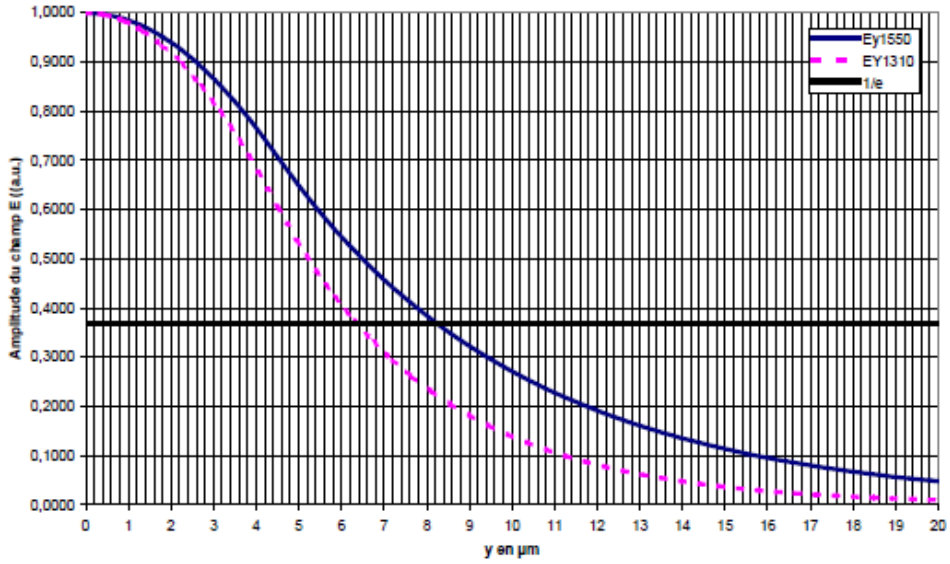
Resulats de l'équation de dispersion pour un guide plan symétrique en TE/TM

Mode 0			Mode 0			Mode 0			Mode 0			Mode 0		
V	U	W	V	U	W	V	U	W	V	U	W	V	U	W
1,3702	0,8766	1,0531	1,4308	0,8950	1,1163	1,4948	0,9134	1,1833	1,5624	0,9318	1,2541	1,6340	0,9502	1,3293
1,3715	0,8770	1,0544	1,4322	0,8954	1,1177	1,4962	0,9138	1,1847	1,5639	0,9322	1,2557	1,6356	0,9506	1,3310
1,3728	0,8774	1,0558	1,4335	0,8958	1,1192	1,4976	0,9142	1,1862	1,5654	0,9326	1,2573	1,6372	0,9510	1,3327
1,3740	0,8778	1,0571	1,4349	0,8962	1,1206	1,4991	0,9146	1,1877	1,5669	0,9330	1,2589	1,6388	0,9514	1,3344
1,3753	0,8782	1,0584	1,4362	0,8966	1,1220	1,5005	0,9150	1,1893	1,5685	0,9334	1,2605	1,6404	0,9518	1,3361
1,3766	0,8786	1,0598	1,4376	0,8970	1,1234	1,5020	0,9154	1,1908	1,5700	0,9338	1,2621	1,6420	0,9522	1,3378
1,3779	0,8790	1,0611	1,4390	0,8974	1,1248	1,5034	0,9158	1,1923	1,5715	0,9342	1,2637	1,6436	0,9526	1,3394
1,3792	0,8794	1,0625	1,4403	0,8978	1,1263	1,5048	0,9162	1,1938	1,5730	0,9346	1,2653	1,6453	0,9530	1,3411
1,3805	0,8798	1,0638	1,4417	0,8982	1,1277	1,5063	0,9166	1,1953	1,5746	0,9350	1,2669	1,6469	0,9534	1,3428
1,3818	0,8802	1,0652	1,4431	0,8986	1,1291	1,5077	0,9170	1,1968	1,5761	0,9354	1,2685	1,6485	0,9538	1,3445
1,3831	0,8806	1,0665	1,4444	0,8990	1,1306	1,5092	0,9174	1,1983	1,5776	0,9358	1,2701	1,6501	0,9542	1,3463
1,3844	0,8810	1,0679	1,4458	0,8994	1,1320	1,5106	0,9178	1,1998	1,5791	0,9362	1,2717	1,6517	0,9546	1,3480
1,3857	0,8814	1,0692	1,4472	0,8998	1,1334	1,5121	0,9182	1,2013	1,5807	0,9366	1,2733	1,6534	0,9550	1,3497
1,3870	0,8818	1,0706	1,4485	0,9002	1,1349	1,5135	0,9186	1,2029	1,5822	0,9370	1,2749	1,6550	0,9554	1,3514
1,3883	0,8822	1,0720	1,4499	0,9006	1,1363	1,5150	0,9190	1,2044	1,5838	0,9374	1,2765	1,6566	0,9558	1,3531
1,3896	0,8826	1,0733	1,4513	0,9010	1,1377	1,5164	0,9194	1,2059	1,5853	0,9378	1,2782	1,6583	0,9562	1,3548
1,3909	0,8830	1,0747	1,4527	0,9014	1,1392	1,5179	0,9198	1,2074	1,5868	0,9382	1,2798	1,6599	0,9566	1,3565
1,3922	0,8834	1,0760	1,4540	0,9018	1,1406	1,5193	0,9202	1,2090	1,5884	0,9386	1,2814	1,6615	0,9570	1,3582
1,3935	0,8838	1,0774	1,4554	0,9022	1,1421	1,5208	0,9206	1,2105	1,5899	0,9390	1,2830	1,6632	0,9574	1,3600
1,3948	0,8842	1,0788	1,4568	0,9026	1,1435	1,5223	0,9210	1,2120	1,5915	0,9394	1,2846	1,6648	0,9578	1,3617
1,3962	0,8846	1,0801	1,4582	0,9030	1,1450	1,5237	0,9214	1,2136	1,5930	0,9398	1,2863	1,6665	0,9582	1,3634
1,3975	0,8850	1,0815	1,4596	0,9034	1,1464	1,5252	0,9218	1,2151	1,5946	0,9402	1,2879	1,6681	0,9586	1,3652
1,3988	0,8854	1,0829	1,4610	0,9038	1,1479	1,5266	0,9222	1,2166	1,5961	0,9406	1,2895	1,6697	0,9590	1,3669
1,4001	0,8858	1,0843	1,4624	0,9042	1,1493	1,5281	0,9226	1,2182	1,5977	0,9410	1,2912	1,6714	0,9594	1,3686
1,4014	0,8862	1,0856	1,4638	0,9046	1,1508	1,5296	0,9230	1,2197	1,5992	0,9414	1,2928	1,6730	0,9598	1,3704
1,4027	0,8866	1,0870	1,4651	0,9050	1,1522	1,5311	0,9234	1,2213	1,6008	0,9418	1,2944	1,6747	0,9602	1,3721
1,4041	0,8870	1,0884	1,4665	0,9054	1,1537	1,5325	0,9238	1,2228	1,6024	0,9422	1,2961	1,6764	0,9606	1,3738
1,4054	0,8874	1,0898	1,4679	0,9058	1,1551	1,5340	0,9242	1,2244	1,6039	0,9426	1,2977	1,6780	0,9610	1,3756
1,4067	0,8878	1,0912	1,4693	0,9062	1,1566	1,5355	0,9246	1,2259	1,6055	0,9430	1,2994	1,6797	0,9614	1,3773
1,4080	0,8882	1,0925	1,4707	0,9066	1,1581	1,5370	0,9250	1,2275	1,6070	0,9434	1,3010	1,6813	0,9618	1,3791
1,4094	0,8886	1,0939	1,4721	0,9070	1,1595	1,5384	0,9254	1,2290	1,6086	0,9438	1,3026	1,6830	0,9622	1,3808
1,4107	0,8890	1,0953	1,4735	0,9074	1,1610	1,5399	0,9258	1,2306	1,6102	0,9442	1,3043	1,6847	0,9626	1,3826
1,4120	0,8894	1,0967	1,4749	0,9078	1,1625	1,5414	0,9262	1,2321	1,6118	0,9446	1,3059	1,6863	0,9630	1,3843
1,4134	0,8898	1,0981	1,4763	0,9082	1,1640	1,5429	0,9266	1,2337	1,6133	0,9450	1,3076	1,6880	0,9634	1,3861
1,4147	0,8902	1,0995	1,4778	0,9086	1,1654	1,5444	0,9270	1,2352	1,6149	0,9454	1,3093	1,6897	0,9638	1,3878
1,4160	0,8906	1,1009	1,4792	0,9090	1,1669	1,5459	0,9274	1,2368	1,6165	0,9458	1,3109	1,6914	0,9642	1,3896
1,4174	0,8910	1,1023	1,4806	0,9094	1,1684	1,5474	0,9278	1,2384	1,6181	0,9462	1,3126	1,6930	0,9646	1,3914
1,4187	0,8914	1,1037	1,4820	0,9098	1,1699	1,5489	0,9282	1,2399	1,6197	0,9466	1,3142	1,6947	0,9650	1,3931
1,4200	0,8918	1,1051	1,4834	0,9102	1,1713	1,5504	0,9286	1,2415	1,6212	0,9470	1,3159	1,6964	0,9654	1,3949
1,4214	0,8922	1,1065	1,4848	0,9106	1,1728	1,5519	0,9290	1,2431	1,6228	0,9474	1,3176	1,6981	0,9658	1,3967
1,4227	0,8926	1,1079	1,4862	0,9110	1,1743	1,5534	0,9294	1,2446	1,6244	0,9478	1,3192	1,6998	0,9662	1,3985
1,4241	0,8930	1,1093	1,4877	0,9114	1,1758	1,5549	0,9298	1,2462	1,6260	0,9482	1,3209	1,7015	0,9666	1,4002
1,4254	0,8934	1,1107	1,4891	0,9118	1,1773	1,5564	0,9302	1,2478	1,6276	0,9486	1,3226	1,7031	0,9670	1,4020
1,4268	0,8938	1,1121	1,4905	0,9122	1,1788	1,5579	0,9306	1,2494	1,6292	0,9490	1,3243	1,7048	0,9674	1,4038
1,4281	0,8942	1,1135	1,4919	0,9126	1,1803	1,5594	0,9310	1,2510	1,6308	0,9494	1,3259	1,7065	0,9678	1,4056
1,4295	0,8946	1,1149	1,4934	0,9130	1,1818	1,5609	0,9314	1,2525	1,6324	0,9498	1,3276	1,7082	0,9682	1,4074

APPENDIX B : Shape of the electri field for our waveguide



carte de champ dans la direction horizontale



Spécifications of the SMF28 fiber

Cable Cutoff Wavelength (λ_{ccf})

$\lambda_{ccf} \leq 1260$ nm

Mode-Field Diameter

Wavelength (nm)	MFD (µm)
1310	9.2 ± 0.4
1550	10.4 ± 0.5